Basic Relationships Between Pixels

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Neighbors of a Pixel

4-Neighbors

A pixel p at coordinates (x, y) has four horizontal and vertical neighbors:

$$(x+1, y),\ (x-1, y),\ (x, y+1),\ (x, y-1)$$

This set is denoted by $N_4(p)$.

Diagonal Neighbors

Coordinates:

$$(x+1, y+1),\ (x+1, y-1),\ (x-1, y+1),\ (x-1, y-1)$$

This set is denoted $N_D(p)$.

8-Neighbors

$$N_8(p) = N_4(p) \cup N_D(p)$$

Adjacency Types

Introduction

Let V be the set of intensity values used to define adjacency.

- In a **binary image**, V often refers to pixels with value 1.
- In a **gray-scale image**, V may contain multiple intensity levels.
- For example, if intensities range from 0 to 255, V can be any subset of these 256 values.

We consider three main types of adjacency:

4-Adjacency

Two pixels p and q are 4-adjacent if $q \in N_4(p)$ and both have values in V.

8-Adjacency

Two pixels p and q are 8-adjacent if $q \in N_8(p)$ and both have values in V.

m-Adjacency

Two pixels p and q are m-adjacent if:

1. $q \in N_4(p)$, or
2. $q \in N_D(p)$ and $N_4(p) \cap N_4(q)$ has no pixels from V.

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Fig: An arrangement of pixels

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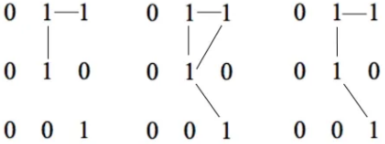

Fig: 4-connectivity of pixels

Fig: 8-connectivity of pixels

Fig: m-connectivity of pixels

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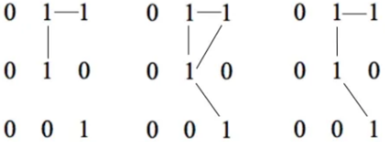

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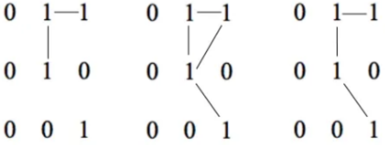

Fig: 4-connectivity of pixels **Fig: 8-connectivity of pixels** **Fig: m-connectivity of pixels**

Why m-Adjacency?

Eliminates ambiguity present in 8-adjacency.

Connectivity, Regions, and Boundaries

Connectivity

Two pixels p and q in set S are connected if there exists a path (4-, 8-, or m-path) between them consisting entirely of pixels in S.

Regions

A **region** is a connected set of pixels. Two regions R_i and R_j are adjacent if their union is connected.

$$\left.\begin{matrix}1&1&1\\1&0&1\\0&1&0\end{matrix}\right\}R_i$$

$$\left.\begin{matrix}0&0&1\\1&1&1\\1&1&1\end{matrix}\right\}R_j$$

Boundary of a Region

$$\begin{matrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{matrix}$$

Boundary

Set of pixels in a region R that have at least one neighbor in R^c (complement of R).

$$\begin{matrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{matrix}$$

Inner vs Outer Boundary

- Inner boundary: within the region.
- Outer boundary: corresponding border in background.

Boundaries vs Edges

Boundary

Global concept: closed path around a region.

Edge

Local concept: set of pixels where intensity derivative exceeds a threshold.

Special Case

In binary images, edges often coincide with region boundaries.

Distance Measures

Definition of a Distance Function (Metric)

For pixels p, q, z with coordinates $(x, y), (s, t), (v, w)$, D is a distance metric if:

1. $D(p, q) \geq 0 \quad$ (and $D(p, q) = 0$ iff $p = q$)
2. $D(p, q) = D(q, p) \quad$ (symmetry)
3. $D(p, z) \leq D(p, q) + D(q, z) \quad$ (triangle inequality)

Euclidean Distance

Formula

$$D_E(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$$

Shape

Points with $D_E \leq r$ form a **disk** of radius r centered at (x, y).

Example

- Coordiantes of $p(x, y) = (0, 0)$ and
- Coordinates of $q(s, t) = (4, 4)$
- $D_e = \sqrt{(4 - 0)^2 + (4 - 0)^2}$
- $D_e = 5.65\, Units$

2(p)	3	2	6	1
6	2	3	6	2
5	3	2	3	5
2	4	3	5	2
4	5	2	3	6(q)

City-Block Distance (D4)

Formula

$$D_4(p, q) = |x - s| + |y - t|$$

Shape

Points with $D_4 \leq r$ form a **diamond** centered at (x, y). Pixels with $D_4 = 1$ are the 4-neighbors of (x, y).

$$\begin{array}{ccccc} & & 2 & & \\ & 2 & 1 & 2 & \\ 2 & 1 & 0 & 1 & 2 \\ & 2 & 1 & 2 & \\ & & 2 & & \end{array}$$

Chessboard Distance (D8)

Formula

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

Shape

Points with $D_8 \leq r$ form a **square** centered at (x, y). Pixels with $D_8 = 1$ are the 8-neighbors of (x, y).

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Coordinate-Based vs Path-Based Distance

Observation

D_m, D_4, and D_8 are independent of paths. However, if m-adjacency is used, the distance is defined as the **shortest m-path** between points.

Implication

The m-distance depends on:

- Values of the pixels along the path.
- Values of their neighbors.

Example Video: Click here